Weighted-Proj quadratic-support local ring

2026-10-28 · proof-boundary observation record · no new target measurements

The checked `WeightedProjQuadraticPoint` module constructs the actual weighted-Proj image of an irreducible support satisfying `u | v^2 - f` and proves its affine chart ideal maximal. `WeightedProjQuadraticLocal` proves that its projective stalk is Noetherian and, when the quintic is separable, a domain. `WeightedProjQuadraticJacobian` now proves that, when 2 is invertible and the quintic is separable, either `2v(root u)` or `f'(root u)` is nonzero. The kernel of the formal tangent row `-f'(root u) dx + 2v(root u) dy` has dimension one over the residue field. The cotangent-rank-one hypothesis for the actual stalk has not been proved.

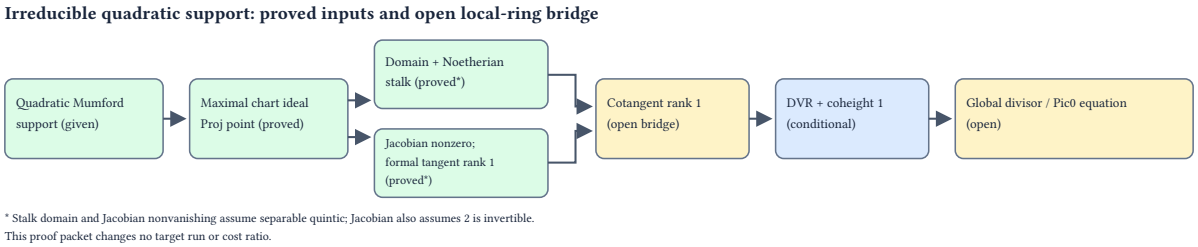

Figure 1: Quadratic-support path. Green inputs compile; yellow is an open bridge; blue requires cotangent rank one.

Pinned Mathlib's DVR criterion turns cotangent dimension one into a DVR at this local ring; `WeightedProjCoheight` then gives coheight one. The cotangent dimension, a global closed-point theorem, principal divisor, and Pic0 class equation remain separate. No target data or cost ratio changed.

An unproved next route is to identify the evaluation kernel with the ideal generated by `u(x)` and `y-v(x)`, then identify the formal tangent kernel with the dual of the localized cotangent space. At nonbranch points, `(y-v)(y+v)=f-v^2` suggests `u(x)` as a local parameter when `y+v` is a unit. The branch case requires its own parameter argument. No quotient/localization cotangent calculation is compiled here. These are proof obligations, not results.

Sources: Mathlib commit `d13f23b723b8a846827a245b89c10fc7d3f11612`, `FieldTheory/Separable.lean`, `AlgebraicGeometry/FunctionField.lean`, `AlgebraicGeometry/Noetherian.lean`, and `RingTheory/DiscreteValuationRing/TFAE.lean`; Stacks Tags 01HD, 00PD, and 04QM; repository `WeightedProjSmoothPoints.lean`, `WeightedProjIsIntegral.lean`, and `WeightedProjLocallyNoetherian.lean`.

The full cohort cost graphs, rows, and ratios are in `research/four_followups_review_20261022/`. This report changes none of those results.